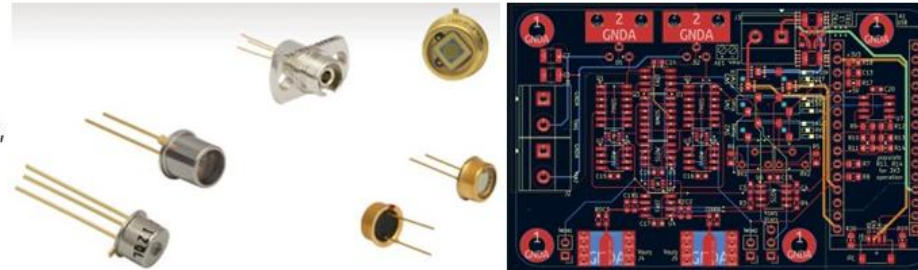
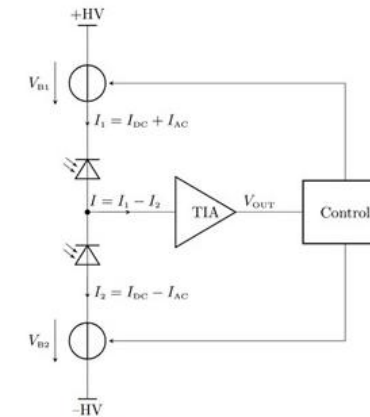


MPP project ideas

Development of a Versatile Auto-Balanced Photodetector

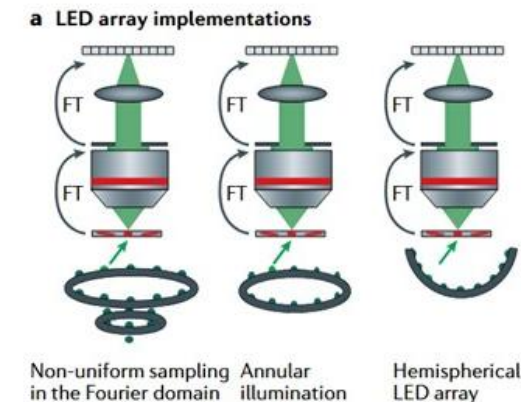
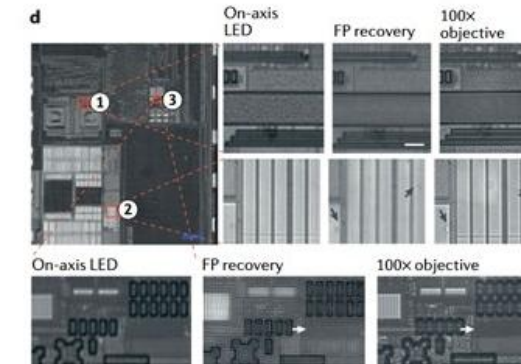
- High speed photodetectors based either on PIN diodes or APDs are one of the common building blocks of electrooptical systems. Due to the numerous different use cases in our labs, each with its own requirements, a versatile and adaptable differential photodetector circuit would certainly be useful. Differential photodetectors are preferably used to detect small AC signals riding on large (common-mode) DC components. In (auto-)balanced photodetectors, this DC component is suppressed to allow the use of high-gain amplifiers, without saturating them [1]. The development of an auto-balanced detector entails:
 - High-frequency analog circuit design (~200 MHz)
 - Converter design for high-voltage biasing (~150 Vdc) of avalanche photodiodes (APD)
 - Controller design for auto-balancing
 - Mechanical design of an enclosure and fixtures for optical tables
- Tasks:
 - Literature research
 - Circuit and PCB design, simulation
 - Fabrication, construction
 - Experimental evaluation (CMRR, transfer characteristic, noise, stability, etc.)
- Supervisors:
 - Péter Vass, peter.vass@tuwien.ac.at
 - Maximilian Thier, maximilian.thier@tuwien.ac.at
- Number of students: 1-2
- References: [1] https://www.rp-photonics.com/balanced_photodetection.html



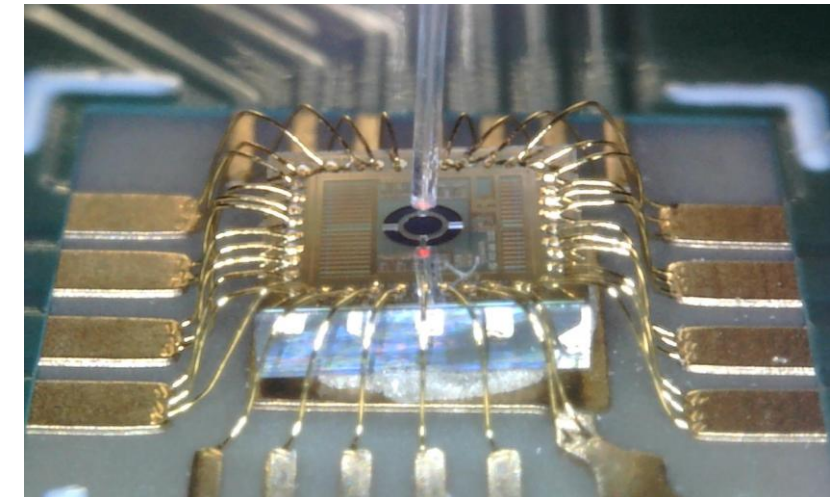
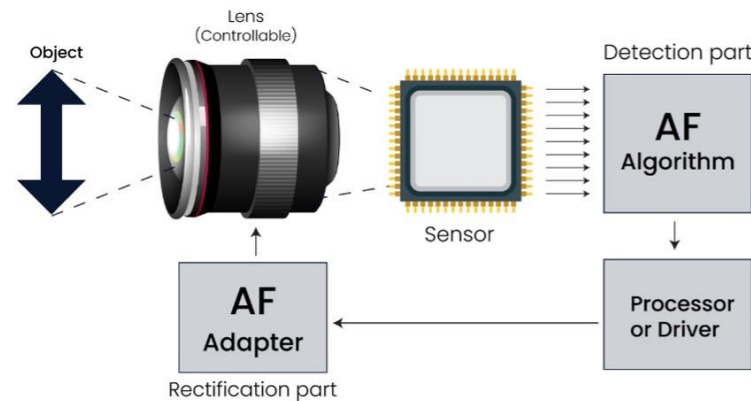
Low-cost Computational Microscopy for the Inspection of Integrated Circuits



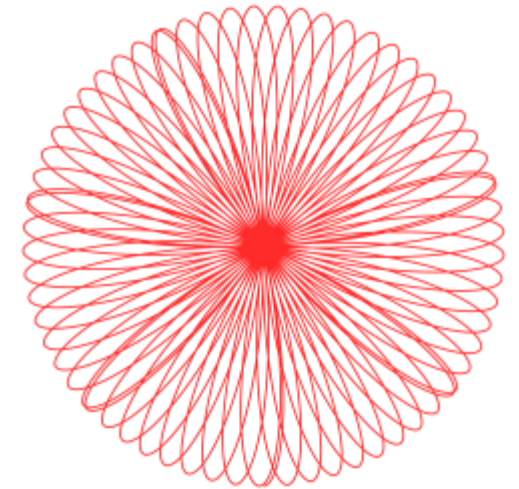
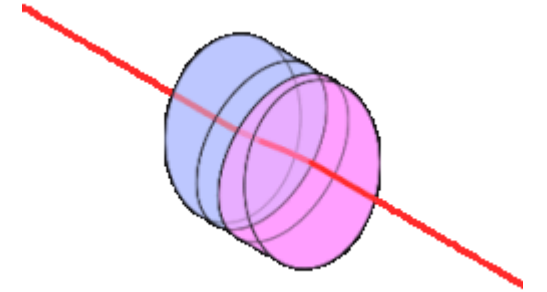
- Fourier ptychographic microscopy (FPM) emerged as a prominent imaging technique in 2013, attracting significant interest due to its remarkable features such as precise phase retrieval, expansive field of view (FOV), and superior resolution. Over the past decade, FPM has become an essential tool in microscopy, with applications in metrology, scientific research, biomedicine, and inspection. [1] . FPM can extend the depth of field by two orders of magnitude compared with a conventional microscope with a similar NA. FPM can also be used to recover an image that is all in focus by digitally refocusing many small segments of the entire image, which are then stitched together. [2] Recently, numerous low-cost systems based on addressable LED arrays have been reported in literature, which showed an impressive improvement of the capabilities of conventional laboratory instruments or consumer grade products with little to no added cost. [3] [4] In this project, such a low-cost method should be implemented or adapted and applied to the imaging of integrated circuits developed by our lab.
- Tasks:
 - Literature research
 - Development of the illumination solution (HW, SW, mechanical)
 - Implementation of the reconstruction algorithm, possibly the adaptation of existing codes [5]
 - Experimental evaluation with real IC samples
- Supervisors:
 - Péter Vass, peter.vass@tuwien.ac.at
 - Maximilian Thier, maximilian.thier@tuwien.ac.at
- Number of students: 1-2
- References:
 - [1] F. Xu et al., "Fourier Ptychographic Microscopy 10 Years on: A Review," Cells, vol. 13, no. 4, p. 324, Feb. 2024, doi: 10.3390/cells13040324.
 - [2] G. Zheng, C. Shen, S. Jiang, P. Song, and C. Yang, "Concept, implementations and applications of Fourier ptychography," Nat Rev Phys, vol. 3, no. 3, pp. 207–223, Mar. 2021, doi: 10.1038/s42254-021-00280-y.
 - [3] G. Zheng, R. Horstmeyer, and C. Yang, "Wide-field, high-resolution Fourier ptychographic microscopy," Nature Photon, vol. 7, no. 9, pp. 739–745, Sep. 2013, doi: 10.1038/nphoton.2013.187.
 - [4] T. Aidukas, R. Eckert, A. R. Harvey, L. Waller, and P. C. Konda, "Low-cost, sub-micron resolution, wide-field computational microscopy using opensource hardware," Sci Rep, vol. 9, no. 1, p. 7457, May 2019, doi: 10.1038/s41598-019-43845-9.
 - [5] <https://github.com/PtyLab/PtyLab.py>



- Mechanical design of autofocus adapter
- 3D printing of required components
- Some electrical design (motor driver and control, position feedback)
- Programming: realization of the real-time autofocus algorithm based on camera images

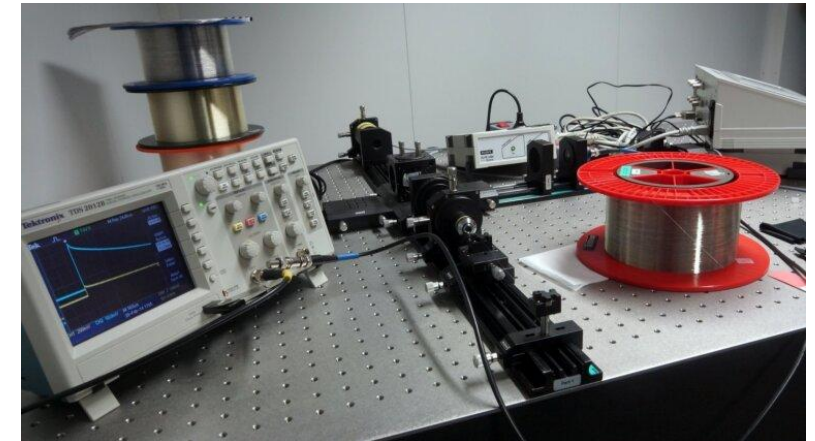
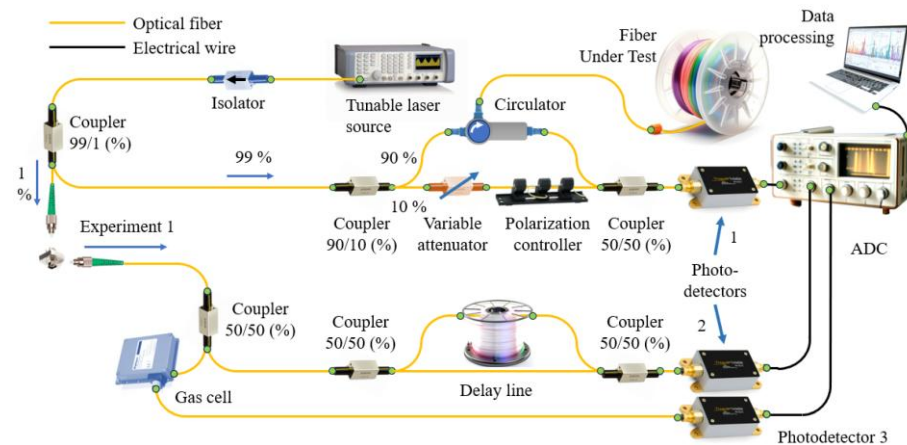
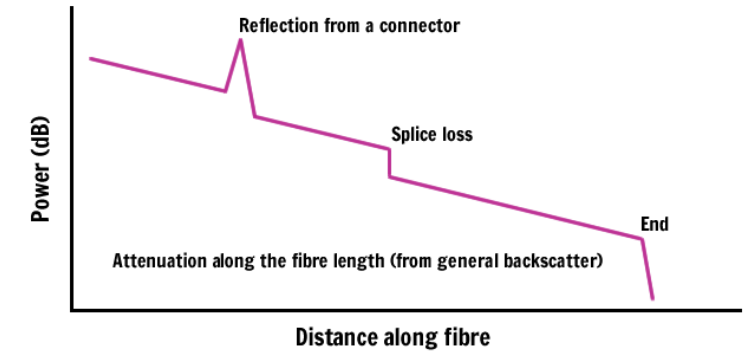


- Mechanical design
- 3D printing
- Some electrical design (motor driver and control, position feedback)



- Optical lab work: build fiber optic OTDR/OFDR setup
- Programming:
 - OTDR/OFDR trace evaluation
 - Measurement automation
- Extension possibility: FPGA implementation

A simple example of how an OTDR trace might look, and how events on the graph can be used to infer the location of link components.



ShapeShifter



■ Description

- Ferrofluid can change its shape depending on the magnetic field in which it is embedded. We want to investigate how well this shape can be controlled, assuming a high resolution control of the magnetic field using a planar coil array embedded into a PCB.

■ Task asked

- Design PCB based coil arrays with 10×10 coil elements and different geometries using signal traces.
- Design a driving PCB with 100 DAC channels capable of driving up to 100mA / channel (not all channels will be active simultaneously)
- Perform tests with a ferrofluid

■ Supervisor

- Michael Hofbauer: michael.hofbauer@tuwien.ac.at

■ Number of students: 1-2



Image generated with ChatGPT



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- Mechanical design of the beam profiler
- 3D printing of the required parts
- Electronic design of motor controller, position sensing
- Programming: automation of the measurement, reconstruction of beam profile from the measured power

